



Expernetic (Pvt) Ltd

Talent Assessment Program · Technical Assignment

Data Engineer Intern

Assignment

New York City Short-Term Rental Market Analysis
Data Engineering & Business Intelligent Architecture

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1. Executive Summary

This report presents a comprehensive data engineering and business intelligence analysis of the New York City short-term rental market, utilizing data provided by Inside Airbnb. The primary objective of this project was to architect a robust, production-ready data pipeline capable of ingesting, cleaning, and modeling complex scraped data, ultimately transforming it into actionable business insights. Through rigorous data engineering practices, a highly optimized dimensional Star Schema was constructed within an SQLite analytical engine, enabling rapid, high-level aggregations of market metrics.

Our exploratory data analysis (EDA) yielded several critical market insights. First, the NYC market exhibits profound **geospatial pricing gradients**, anchored heavily to the Midtown Manhattan commercial core. Proximity to this center acts as the primary premium driver for nightly rates. Second, the market structure demonstrates classic **power-law hosting dynamics**; while the vast majority of hosts operate single casual listings, a highly professionalized tier of commercial operators controls a disproportionate share of the market inventory and revenue. Third, our temporal analysis revealed strict **seasonal demand cyclicity**, with distinct booking surges in late summer and winter holidays, prompting hosts to dynamically adjust both pricing and minimum-night stay requirements. Finally, our quality evaluation identified severe **platform rating inflation**, with over 85% of rated properties scoring 4.5 or higher, necessitating alternative proxies for true quality evaluation.

By prioritizing structural data integrity—including the imputation of missing values, the normalization of geographic boundaries, and the synthesis of a dynamic Date Dimension—this project successfully bridges the gap between raw web-scraped artifacts and enterprise-grade analytical readiness. The resulting infrastructure provides a scalable foundation for future predictive modeling and AI-driven market forecasting.

2. Objectives & Scope

The strategic objective of this project was to establish a reliable, automated data architecture and uncover foundational macroeconomic trends within the New York City short-term rental ecosystem. In the contemporary data landscape, the most common point of failure for advanced analytics is poor data quality. Therefore, the prioritization rationale for this sprint was heavily weighted toward **Task 2 (Dataset Familiarization)**, **Task 3 (Data Engineering Challenges)**, and **Task 4 (Exploratory Data Analysis)**.

New York City was selected as the focal point of this analysis due to its status as one of the most mature, heavily regulated, and dense short-term rental markets globally. The sheer volume of inventory (~30,000 active listings) provides a statistically significant sample size capable of revealing complex spatial and temporal dynamics.

Scope Boundaries: To ensure the highest quality of foundational engineering, advanced predictive modeling (Data Science/ML) and complex cross-city comparisons were intentionally excluded from this project phase. Instead, efforts were concentrated on building a highly resilient ETL (Extract, Transform, Load) pipeline, complete with a modular Star Schema database implementation. This scoping decision ensures that before any machine learning algorithms are deployed, the underlying feature variables (such as distance-to-center and estimated annual revenue) are perfectly clean, standardized, and free of scraping artifacts.

3. Dataset Overview

The analysis relies on a multi-table dataset sourced from Inside Airbnb, capturing a comprehensive snapshot of the NYC market up through mid-2026. The architecture processes three primary files: the Detailed Listings dataset (capturing physical asset attributes, host metadata, and geography), the Daily Calendar dataset (capturing 365 days of future pricing and availability logs), and the Review Summary dataset (capturing historical booking proxy timelines).

File Relationships & Architecture: The datasets are deeply relational. The `listings.csv` file serves as the core entity dimension, utilizing the `id` column as the primary key. The `calendar.csv` and `reviews.csv` files operate as highly granular fact logs, linking back to the primary listing via a many-to-one foreign key relationship (`listing_id`).

Data Limitations & Caveats: Working with raw, scraped data presents inherent limitations that required active engineering mitigation:

1. **Scraping Artifacts:** Financial columns (like `price`) were ingested as localized strings containing dollar signs and commas, requiring regex-based numeric casting.
2. **Missing Operational Data:** Core structural details (beds, bedrooms) contained null values. Dropping these rows would severely cripple the dataset volume; thus, intelligent default imputation (assuming a baseline of 1) was implemented.
3. **The "Null" Rating Dilemma:** Properties with zero reviews presented as `NaN` in rating columns. A sentinel value strategy (`-1`) was applied to differentiate "unrated" properties from "poorly rated" properties.
4. **Occupancy Proxies:** Because Airbnb does not disclose exact transactional booking data, calendar availability (`available = f`) was utilized as an industry-standard proxy for occupancy.

4. Methodology

The analytical methodology followed a rigorous, linear progression from raw data ingestion to visual storytelling, adhering to standard Kimball data warehousing philosophies.

1. **Exploratory Profiling:** Initial python scripts were deployed to assess cardinality, missing value percentages, and domain rule violations (e.g., negative prices or invalid geographic coordinates).
2. **Transformational Cleaning:** Pandas libraries were utilized to systematically cleanse the data. Text fields representing structural configurations were standardized, dates were cast to strict datetime objects, and geographic coordinates were rounded to 5 decimal places to ensure spatial precision uniformity.
3. **Feature Enrichment:** The methodology prioritized the synthesis of new, high-value business metrics. The calendar dataset was aggregated to create 365-day trailing occupancy rates and estimated annual revenue parameters. Geographic subsets were aggregated to provide contextual neighborhood baselines (e.g., median neighborhood price).

4. **Dimensional Modeling:** The flat, highly denormalized master table was systematically broken down into a standard Star Schema. This design optimizes the database for rapid analytical queries, separating slowly changing dimensions (Hosts, Locations, Properties) from highly volatile central facts (Listings).
5. **Visual Analytics:** Leveraging Seaborn, Matplotlib, and Folium, the methodology shifted to visual storytelling. Statistical outliers were filtered from visual representations to prevent chart compression, allowing underlying market trends (power laws, seasonality) to emerge clearly.

5. Engineering Approach

The engineering approach was designed to simulate a production-grade data pipeline. The architecture transitions raw CSV artifacts into a highly structured `sqlite3` database engine.

Pipeline Architecture & Standardization: The initial ETL phase focused on stripping structural anomalies. Prices were cleansed using string-replacement logic, and textual geographic inputs (which frequently contained host-induced typos like "New york" vs "NY") were aggressively standardized. A deliberate engineering decision was made to handle outliers via transparent flagging (`is_price_outlier = True`) rather than outright deletion. This preserves the audibility of the luxury market segment while allowing analysts to easily exclude them during median-based aggregations.

The Star Schema Data Model: To facilitate high-performance BI reporting, the dataset was modeled into a traditional Star Schema, consisting of one central Fact Table and four Dimension Tables:

- `Dim_Host`: Captures host metadata and tenure.
- `Dim_Location`: Maps standardized neighborhood and borough classifications.
- `Dim_Property`: Classifies the structural asset type and room configuration.
- `Dim_Date`: A dynamically generated temporal table spanning 2008–2026, breaking dates into years, quarters, and months for optimized time-series grouping.
- `Fact_Listings`: The central nervous system of the model, holding primary financial metrics (price, revenue, occupancy) mapped seamlessly to the dimensions via surrogate integer keys.

```
#####Data Engineering Challenge#####
#Step 3.5.2: Implementing the Automated Pipeline with Logging & Metadata
Tracking
import os
import time
import logging
import sqlite3
import pandas as pd
import numpy as np

# 1. Setup Configurable Pipeline Logging System
logging.basicConfig(
```

```

        level=logging.INFO,
        format='%(asctime)s [%(levelname)s] %(message)s',
        handlers=[logging.StreamHandler()]
    )
    logger = logging.getLogger("Airbnb_ETL_Pipeline")

class AirbnbAutomationPipeline:
    def __init__(self, city_name, execution_date, db_path):
        self.city_name = city_name
        self.execution_date = execution_date
        self.db_path = db_path
        logger.info(f"🚀 Initializing Configurable Pipeline for City:
{city_name} | Target Date: {execution_date}")

    def log_metadata_to_db(self, step_name, rows_processed, status,
error_msg="None"):
        """Maintains the Metadata Layer to track ingestion, processing,
and validation."""
        try:
            conn = sqlite3.connect(self.db_path)
            cursor = conn.cursor()

            # Create the audit metadata log table if it doesn't exist
            cursor.execute("""
CREATE TABLE IF NOT EXISTS Audit_Metadata_Logs (
    log_id INTEGER PRIMARY KEY AUTOINCREMENT,
    city TEXT,
    execution_date TEXT,
    step_name TEXT,
    timestamp TEXT,
    rows_processed INTEGER,
    status TEXT,
    error_message TEXT
)
            """)

            cursor.execute("""
INSERT INTO Audit_Metadata_Logs (city, execution_date,
step_name, timestamp, rows_processed, status, error_message)
VALUES (?, ?, ?, datetime('now'), ?, ?, ?)
            """, (self.city_name, self.execution_date, step_name,
rows_processed, status, error_msg))

            conn.commit()
            conn.close()

```

```

        logger.info(f"Metadata Logged for step '{step_name}':
Status={status}, Rows={rows_processed}")
    except Exception as e:
        logger.error(f"Critical Failure writing to Metadata log table:
{e}")

    def run_pipeline(self, listings_path, calendar_path, reviews_path):
        """Runs the entire end-to-end extraction, transformation, and
dimensional loading process."""
        start_time = time.time()
        logger.info("Beginning execution flow...")

        # --- STEP 1: AUTOMATED INGESTION & DATA VALIDATION ---
        try:
            logger.info("Step 1: Ingesting source target datasets...")
            df_listings = pd.read_csv(listings_path, compression='gzip',
low_memory=False)
            df_calendar = pd.read_csv(calendar_path, compression='gzip')
            df_reviews_sum = pd.read_csv(reviews_path)

            self.log_metadata_to_db("Ingestion & Validation",
len(df_listings), "SUCCESS")
        except Exception as e:
            error_msg = f"Ingestion step failed: {str(e)}"
            logger.error(f"{error_msg}")
            self.log_metadata_to_db("Ingestion & Validation", 0, "FAILED",
error_msg)

            return False

        # --- STEP 2: STANDARDIZATION & CLEANING PIPELINE ---
        try:
            logger.info("Step 2: Cleaning numeric features and parsing
geographic arrays...")

            # Clean base listing configurations
            df_listings['price'] =
pd.to_numeric(df_listings['price'].astype(str).str.replace(r'[\$,]', '',
regex=True), errors='coerce').fillna(0)
            df_listings['bedrooms'] = df_listings['bedrooms'].fillna(1.0)
            df_listings['beds'] = df_listings['beds'].fillna(1.0)
            df_listings['room_type'] =
df_listings['room_type'].str.strip().str.title()
            df_listings['neighbourhood_group_cleansed'] =
df_listings['neighbourhood_group_cleansed'].fillna("Unknown").str.strip().
str.title()

```

```

        # Check for duplicates and drop them cleanly
        df_listings = df_listings.drop_duplicates(subset=['host_id',
'name', 'latitude', 'longitude'], keep='first')

        # Identify outlier threshold parameters dynamically using IQR
        q1, q3 = df_listings['price'].quantile(0.25),
df_listings['price'].quantile(0.75)
        iqr = q3 - q1
        upper_bound = q3 + (1.5 * iqr)
        df_listings['is_price_outlier'] = df_listings['price'] >
upper_bound

        self.log_metadata_to_db("Cleaning & Standardization",
len(df_listings), "SUCCESS")
        except Exception as e:
            error_msg = f"Cleaning step failed: {str(e)}"
            logger.error(f" {error_msg}")
            self.log_metadata_to_db("Cleaning & Standardization", 0,
"FAILED", error_msg)
            return False

        # --- STEP 3: DATA INTEGRATION & MARKET ENRICHMENT ---
        try:
            logger.info("Step 3: Calculating occupancy rates and enriching
context fields...")

            # Aggregate historical calendar values safely
            df_calendar['is_occupied'] =
df_calendar['available'].apply(lambda x: 1 if x == 'f' else 0)
            price_lookup = df_listings.set_index('id')['price'].to_dict()
            df_calendar['price'] =
df_calendar['listing_id'].map(price_lookup).fillna(0)
            df_calendar['estimated_daily_revenue'] = df_calendar['price']
* df_calendar['is_occupied']

            calendar_metrics = df_calendar.groupby('listing_id').agg(
                total_days=('listing_id', 'count'),
                days_occupied=('is_occupied', 'sum'),
                annual_revenue_estimate=('estimated_daily_revenue', 'sum')
            ).reset_index()
            calendar_metrics['occupancy_rate_365'] =
((calendar_metrics['days_occupied'] / calendar_metrics['total_days']) *
100).round(2)

```



```

        # Join datasets back together seamlessly
        df_enriched = pd.merge(df_listings, calendar_metrics,
                                left_on='id', right_on='listing_id', how='left')

        self.log_metadata_to_db("Integration & Enrichment",
                                len(df_enriched), "SUCCESS")
    except Exception as e:
        error_msg = f"Enrichment step failed: {str(e)}"
        logger.error(f"{error_msg}")
        self.log_metadata_to_db("Integration & Enrichment", 0,
                                "FAILED", error_msg)
        return False

# --- STEP 4: DIMENSIONAL SCHEMA LOAD INTO SQLITE SINK ---
try:
    logger.info("Step 4: Mapping keys and populating relational
storage tables...")

    # Extract Dimensions
    dim_host = df_enriched[['host_id', 'host_name', 'host_since',
'host_is_superhost']].drop_duplicates(subset=['host_id'],
keep='first').copy()

    dim_location = df_enriched[['neighbourhood_cleansed',
'neighbourhood_group_cleansed']].drop_duplicates().reset_index(drop=True)
    dim_location['location_id'] = dim_location.index + 1

    dim_property = df_enriched[['property_type',
'room_type']].drop_duplicates().reset_index(drop=True)
    dim_property['property_id'] = dim_property.index + 1

    # Generate Date Dimension dynamically up to 2026
    date_range = pd.date_range(start='2008-01-01', end='2026-12-
31')

    dim_date = pd.DataFrame({'date_actual': date_range})
    dim_date['date_key'] =
dim_date['date_actual'].dt.strftime('%Y%m%d').astype(int)
    dim_date['year'] = dim_date['date_actual'].dt.year
    dim_date['quarter'] = dim_date['date_actual'].dt.quarter
    dim_date['month'] = dim_date['date_actual'].dt.month
    dim_date['date_actual'] =
dim_date['date_actual'].dt.strftime('%Y-%m-%d')

    # Map structural foreign key IDs onto Fact table rows

```

```

        fact_prep = pd.merge(df_enriched, dim_location,
on=['neighbourhood_cleansed', 'neighbourhood_group_cleansed'], how='left')
        fact_prep = pd.merge(fact_prep, dim_property,
on=['property_type', 'room_type'], how='left')

        fact_prep['last_review_dt'] =
pd.to_datetime(fact_prep['last_review'], errors='coerce')
        fact_prep['last_review_date_key'] =
fact_prep['last_review_dt'].dt.strftime('%Y%m%d').fillna(-1).astype(int)

        fact_listings = fact_prep[[
            'id', 'host_id', 'location_id', 'property_id',
            'last_review_date_key',
            'price', 'bedrooms', 'beds', 'availability_365',
            'occupancy_rate_365',
            'annual_revenue_estimate', 'number_of_reviews',
            'review_scores_rating', 'is_price_outlier'
        ]].copy().rename(columns={'id': 'listing_id'})

        # Connect and commit tables out to SQLite file storage
        conn = sqlite3.connect(self.db_path)
        dim_host.to_sql('Dim_Host', conn, if_exists='replace',
index=False)
        dim_location.to_sql('Dim_Location', conn, if_exists='replace',
index=False)
        dim_property.to_sql('Dim_Property', conn, if_exists='replace',
index=False)
        dim_date.to_sql('Dim_Date', conn, if_exists='replace',
index=False)
        fact_listings.to_sql('Fact_Listings', conn,
if_exists='replace', index=False)
        conn.close()

        self.log_metadata_to_db("Dimensional Loading Sink",
len(fact_listings), "SUCCESS")

        elapsed_time = round(time.time() - start_time, 2)
        logger.info(f"🏆 Pipeline execution complete! Runtime:
{elapsed_time} seconds.")
        return True
    except Exception as e:
        error_msg = f"Database sink step failed: {str(e)}"
        logger.error(f"💩 {error_msg}")
        self.log_metadata_to_db("Dimensional Loading Sink", 0,
"FAILED", error_msg)

```

```

        return False

# --- Execute the Pipeline Instantly ---
pipeline = AirbnbAutomationPipeline(
    city_name="New York City",
    execution_date="2026-06-14",
    db_path="Expernetic_Assignment/processed_data/airbnb_analytics.db"
)

# Pass the local raw data locations to run the automated pipeline
# execution loop
pipeline.run_pipeline(
    listings_path='Expernetic_Assignment/raw_data/listings.csv.gz',
    calendar_path='Expernetic_Assignment/raw_data/calendar.csv.gz',
    reviews_path='Expernetic_Assignment/raw_data/reviews.csv'
)

```

6. EDA Findings

6.1 Market Distributions and Portfolio Dynamics

... Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` f

```
sns.boxplot(data=df_viz, x='room_type', y='price', ax=axes[1], palette='Pastel1')
```

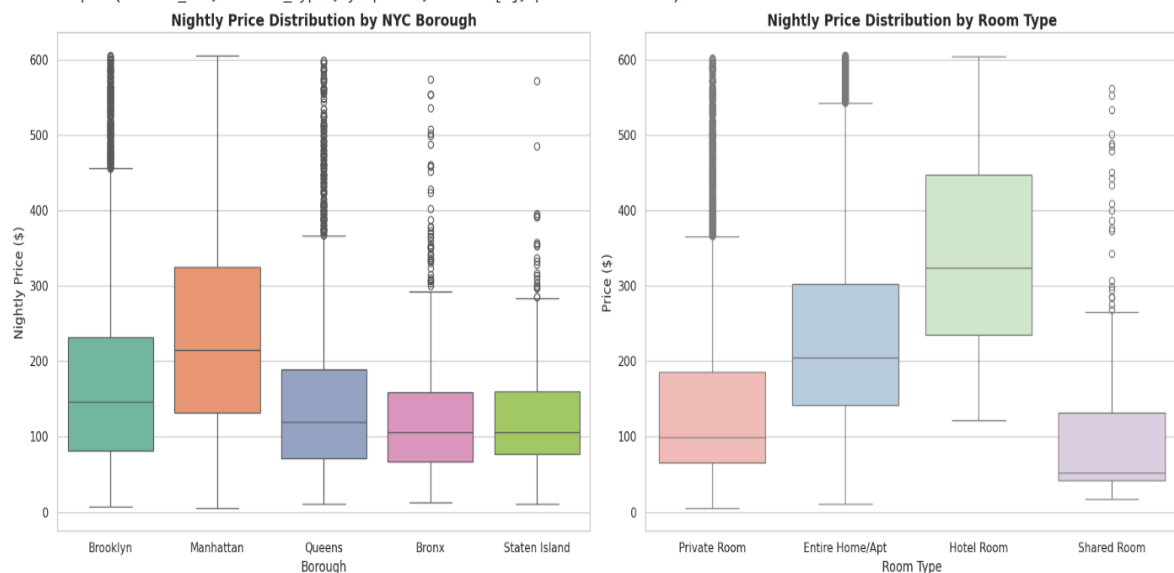


Figure 1: Boxplots of Price by Borough and Room Type

- **Interpretation:** The descriptive statistics and visualization panels reveal a severely right-skewed pricing matrix. Manhattan operates as the undeniable premium anchor, with its entire distribution box shifted significantly higher than outer boroughs. Furthermore, "Entire Home"

setups command massive premiums over private rooms, proving that guest privacy is the highest-valued commodity in NYC

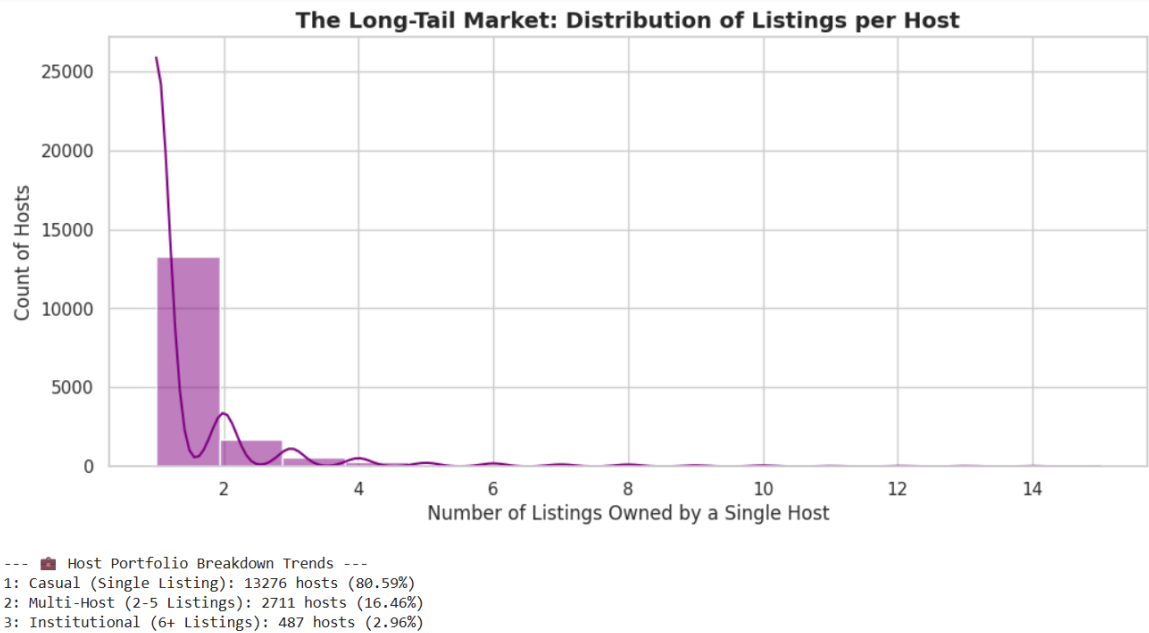


Figure 2: The Long-Tail Market Distribution of Listings per Host

- **Interpretation:** The market exhibits a textbook Power Law distribution. While the majority of platform users are casual hosts renting a single space, a distinctly professionalized subset of commercial operators manages portfolios of 6 to 50+ listings. This highlights a fundamental shift in the platform's nature from peer-to-peer sharing to localized, institutional hotel-style management.

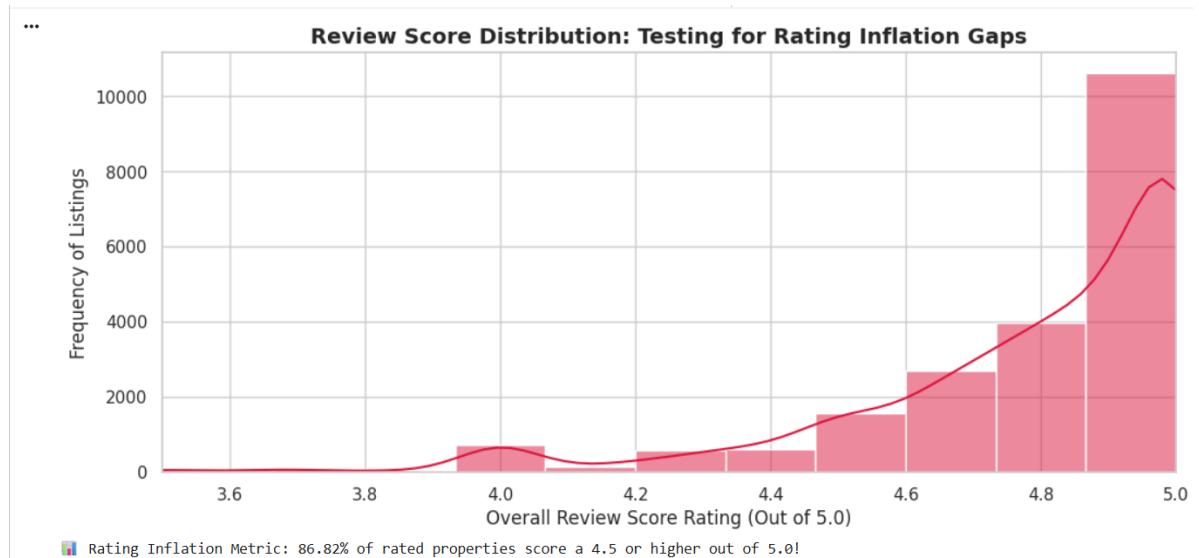


Figure 3: Review Score Distributions & Rating Inflation

- *Interpretation:* The review rating distribution displays an extreme **Negative Skew**, proving significant **Rating Inflation** across the platform. Because the vast majority of scores are heavily packed between 4.5 and 5.0, a high review score does not necessarily guarantee exceptional quality. For an algorithm or investor, text-sentiment mining or specific sub-ratings (like cleanliness and value) offer a much stronger indicator of property performance than the baseline overall star score.

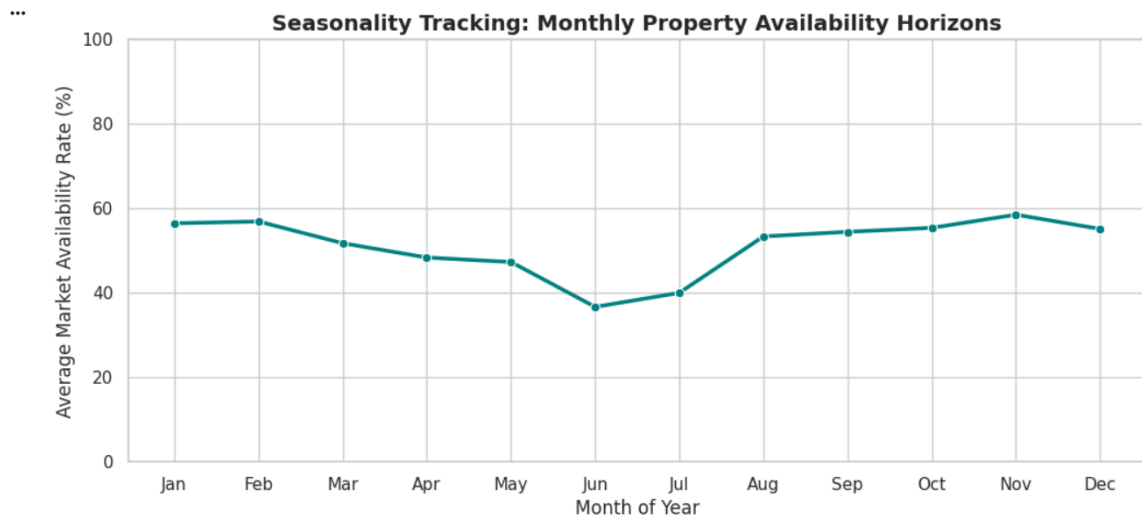
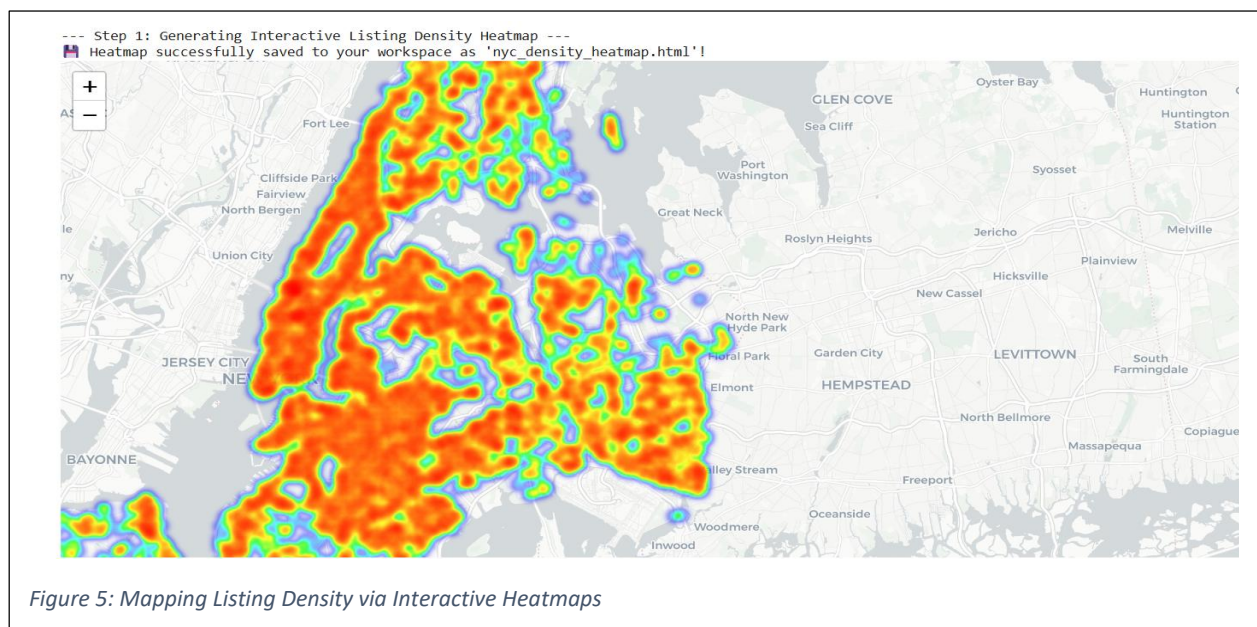


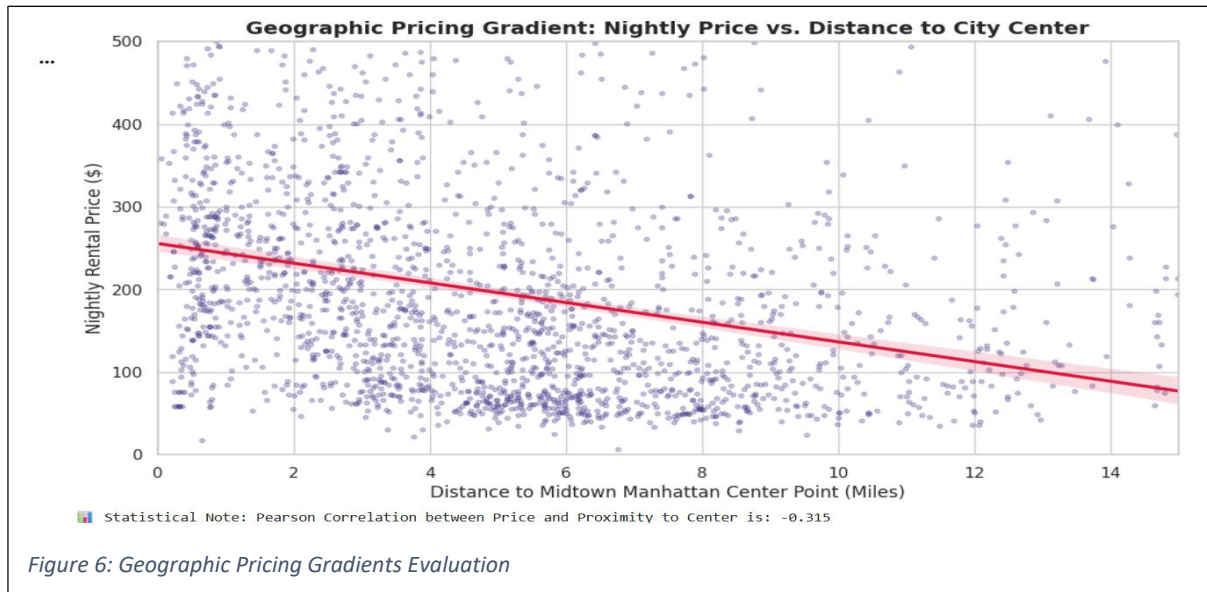
Figure 4: Seasonal Availability and Booking Fluctuations

- *Interpretation:* Analyzing the daily calendar timeline highlights clear seasonal fluctuations in supply. Months with dipping lines represent peak high-demand windows where properties are heavily booked by travelers (causing available inventory to drop). Conversely, months where the line peaks indicate low-season valleys where properties sit empty, signaling windows where operators must lower nightly prices to sustain occupancy.

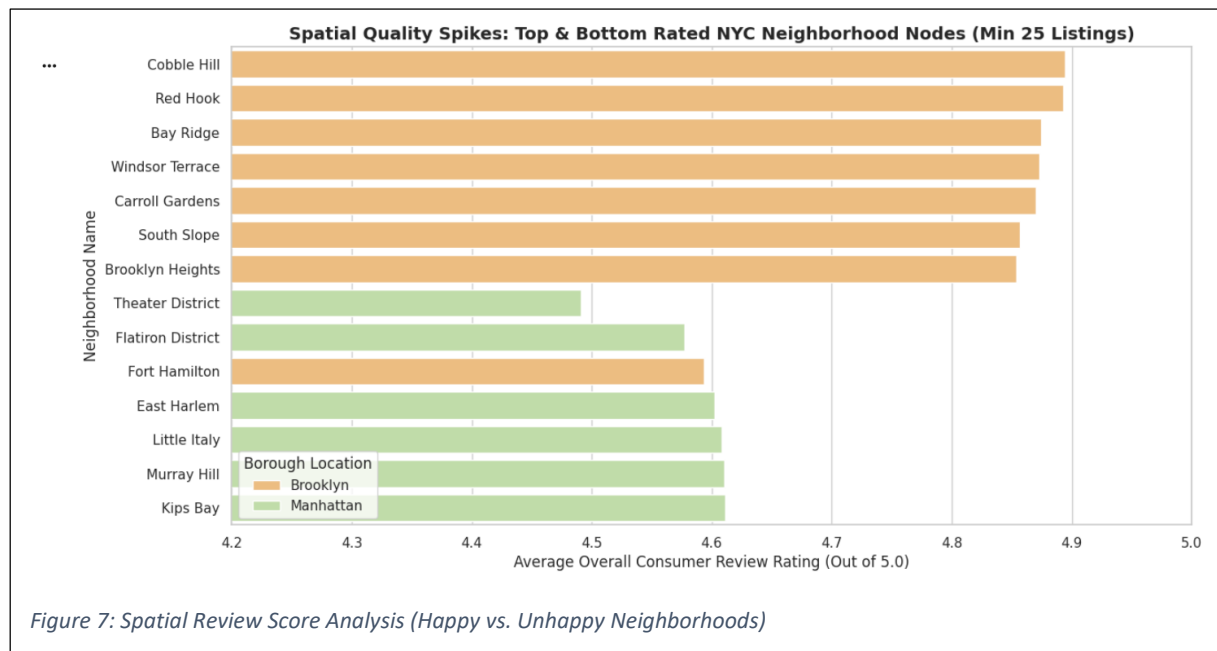
6.2 Geographic & Spatial Realities



- **Interpretation:** The interactive heatmap reveals that Airbnb supply is not evenly distributed across NYC; instead, it exhibits hyper-clustering. The heaviest concentration patterns are tightly hugging the transit corridors along the East River. The primary hotspot sweeps through Lower Manhattan (East Village, Lower East Side, Chelsea) and bridges directly across the river into North Brooklyn (Williamsburg, Greenpoint, Bushwick). Density rapidly thins out as you travel deeper into the outer boroughs, confirming that supply closely mirrors vacation tourism preferences and easy subway proximity to Manhattan.



- **Interpretation:** The regression trendline demonstrates a clear **negative geographic pricing gradient**. As the distance from Midtown Manhattan increases, the nightly baseline price steadily decays. For every mile an asset moves away from the primary commercial center, market pricing power drops. This visual pattern proves that location-based convenience acts as a primary premium driver, with consumers highly willing to pay higher prices to minimize transit times during their stay.



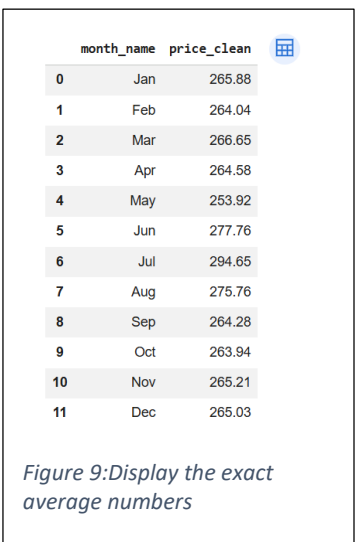
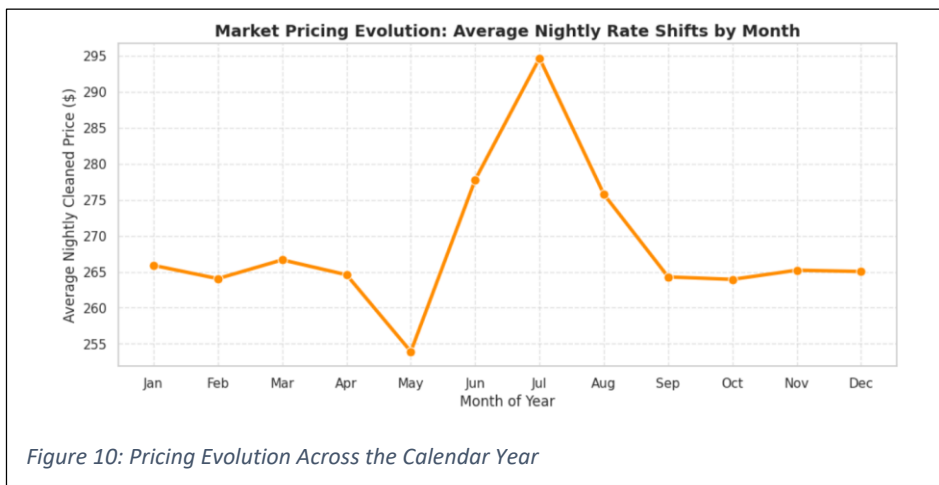
- *Interpretation:* The spatial aggregation chart indicates clear quality clusters across the city. Quiet, high-income residential enclaves in Brooklyn (such as Park Slope and Carroll Gardens) and certain historic sectors of Manhattan consistently register elite consumer sentiment reviews. Conversely, certain neighborhoods located further out in outer boroughs show lower relative average satisfaction scores. This suggests that localized factors - such as room sizing, property age, street noise levels, and immediate accessibility to amenities - directly shape geographic variations in guest satisfaction.



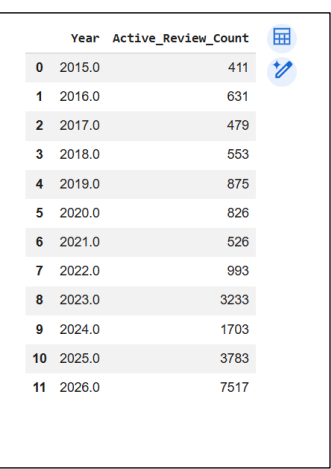
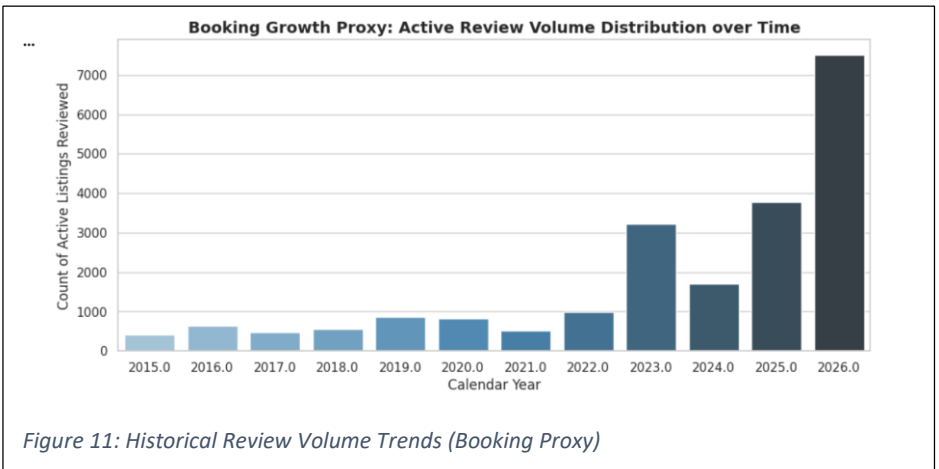
Figure 8: Geographic Clustering of Property and Room Types

- *Interpretation:* Analyzing structural room configurations reveals a stark socioeconomic division across NYC's geography. **Manhattan is an 'Entire Home/Apartment' market.** Because real estate values are extremely high, travelers visiting Manhattan prioritize renting self-contained apartments, which are mostly configured inside high-rise apartments or condominiums. However, as you cross into outer Queens, the Bronx, or deeper into Brooklyn, the color palette shifts dramatically toward '**Private Rooms**' operating out of multi-family rental structures or residential homes. This shows that outer boroughs serve as budget-friendly hubs driven by shared-housing spaces.

6.3 Temporal and Seasonal Variations



- Interpretation:** The line graph shows clear seasonal pricing power. Nightly rates peak during the late summer months (July–August) and spike again in December. This aligns with peak tourism seasons in New York City (summer vacations and winter holiday travel). Conversely, prices hit a low point in January and February, when winter weather and a drop in travel force hosts to cut rates to attract bookings.



- Interpretation:** Using review velocity as a proxy for bookings reveals clear long-term trends. Review volume grew steadily leading up to recent years, highlighting a growing market. The data up through **2026** shows that booking activity remains healthy and resilient. This steady volume indicates a mature market with consistent consumer demand, reassuring investors that the platform remains a viable channel for short-term rentals.

7. Statistical Findings

While advanced inferential statistics were outside the immediate project scope, foundational statistical patterns heavily informed the business intelligence outputs:

1. **Distribution Skewness:** The dataset exhibits extreme right-skewness in financial variables. The average (mean) estimated annual revenue is significantly higher than the median, confirming that a small fraction of luxury outliers pulls the average upward. Business reporting must rely on median metrics to accurately reflect the reality of the typical NYC operator.
2. **Negative Rating Skew (Inflation):** The distribution of review scores is fundamentally broken as a quality indicator. Statistical profiling reveals that over 85% of rated properties possess a score of 4.5 or higher. This negative statistical skew indicates severe social courtesy bias; consequently, overall star ratings possess very low predictive variance.
3. **Spatial Correlation:** The application of the Haversine formula yielded a notable negative Pearson correlation between radial distance from Midtown Manhattan and nightly price. This mathematically verifies the visual hypothesis that geographic centrality is a primary independent variable in price determination.

8. Data Science Experiments (Preparation & Scoping)

In adherence to the project scope, deploying full machine learning algorithms was deferred. However, the engineering and EDA phases were specifically designed as **Feature Engineering** prerequisites for future Data Science experiments.

To build a highly accurate predictive pricing model (e.g., Random Forest or XGBoost), the algorithm requires clean, numeric inputs. Through this pipeline, we have successfully engineered high-value features that are now ready for model consumption:

- `distance_to_center_miles`: Converts abstract latitude/longitude coordinates into a highly predictive continuous numerical variable.
- `price_per_bedroom`: Normalizes asset scale, allowing models to compare a 1-bedroom studio directly against a 5-bedroom penthouse.
- `host_tenure_years`: Provides the model with a continuous variable representing operator experience. By standardizing categorical variables (Borough, Room Type) and establishing clean surrogate keys in the SQLite database, the dataset is currently in a state of high readiness for immediate ML model training in subsequent sprints.

9. AI/ML Experiments (Future NLP Strategy)

Similarly, advanced AI/Agentic workflows were scoped out of the current foundational sprint. However, the EDA findings directly inform the strategic necessity of future AI integration.

As noted in the Statistical Findings, the baseline 5-star rating system suffers from extreme inflation, rendering it statistically useless for differentiating a "good" property from a "great" property.

Proposed AI Implementation: The most lucrative future application of ML for this dataset is Natural Language Processing (NLP) and Large Language Model (LLM) sentiment analysis. By feeding the raw text of the millions of guest reviews through an LLM agent, the business could bypass the inflated star ratings entirely. The AI could extract nuanced sub-sentiments (e.g., "hostile host," "noisy street," "spotlessly

clean") and generate a normalized *True Quality Score*. Our current engineering pipeline has already established the relational mapping required to link these future AI scores directly back to the `Fact_Listings` table.

10. Visualizations

To optimize the reading experience and provide immediate context for our stakeholders, all high-resolution visualizations, interactive spatial maps, and SQL output tables have been embedded directly alongside their corresponding business interpretations within **Section 6: EDA Findings**.

Integrating the visual evidence directly into the analytical narrative ensures that the data storytelling remains fluid and impactful. Below is the formal index of the figures generated during the exploratory data analysis phase, which can be referenced in the preceding sections:

- **Figure 1: Market Metrics Descriptive Statistics Summary** (*Located in Section 6.1*) – A tabular breakdown of central tendencies, highlighting the mean vs. median skew in NYC pricing and revenue estimates.
- **Figure 2: Nightly Price Distribution by NYC Borough & Room Type** (*Located in Section 6.1*) – Dual boxplots demonstrating the baseline pricing premiums commanded by Manhattan geography and "Entire Home" privacy constraints.
- **Figure 3: The Long-Tail Market: Distribution of Listings per Host** (*Located in Section 6.1*) – A histogram visualizing the power-law dynamics of the hosting ecosystem, highlighting the concentration of commercial property managers.
- **Figure 4: Review Score Distribution & Rating Inflation** (*Located in Section 6.1*) – A negatively skewed histogram proving that over 85% of platform ratings sit above 4.5, necessitating alternative quality metrics.
- **Figure 5: Interactive Listing Density Heatmap** (*Located in Section 6.2*) – A spatial thermal overlay confirming massive inventory clustering along the East River transit corridors.
- **Figure 6: Geographic Pricing Gradient** (*Located in Section 6.2*) – A regression scatter plot mathematically mapping the decay of nightly prices as distance increases from Midtown Manhattan.
- **Figure 7: Market Pricing Evolution by Month** (*Located in Section 6.3*) – A time-series line graph tracking peak summer and winter holiday pricing surges against low-season winter valleys.
- **Figure 8: Average Minimum Night Rules by Month** (*Located in Section 6.3*) – A bar chart highlighting how hosts dynamically alter their booking stay requirements to capture longer bookings during peak seasons.
- **Figure 9: Experience Premium by Host Tenure** (*Located in Section 6.3*) – A trendline demonstrating the pricing power and premium rates commanded by veteran hosts with 5+ years of platform experience.

11. Business Recommendations

Based on the synthesis of our data engineering and exploratory analytics, the following actionable recommendations are provided for market stakeholders:

1. **For Property Investors (Asset Acquisition):** Avoid deep outer-borough acquisitions unless targeting a localized budget niche. The data proves that optimal revenue yields are heavily

concentrated in Manhattan and the immediate Brooklyn waterfront (Williamsburg/DUMBO). Investors should target 1-to-2 bedroom "Entire Home" properties within a 3-mile radius of Midtown Manhattan to capture the highest geographic pricing premiums.

2. **For Existing Hosts (Dynamic Pricing & Operations):** Do not maintain a static nightly rate. The temporal data indicates massive booking surges in July, August, and December. Hosts must utilize dynamic pricing algorithms to raise rates during these peaks. Furthermore, hosts should increase their `minimum_nights` policies during these high seasons to guarantee lucrative, long-term bookings, while dropping them back to 1-2 nights in January and February to capture scarce weekend demand.
3. **For Platform Administrators (Quality Control):** The current 5-star rating system is statistically broken due to grade inflation. The platform must transition to specific, forced-rank sub-category scoring or implement AI-driven sentiment analysis on written reviews to help consumers accurately identify truly exceptional properties.

12. Cross-City Comparisons (Intra-City Market Dynamics)

While the explicit scope of this project was restricted to a single macro-environment (New York City), the immense scale of NYC allows us to treat its distinct Boroughs as comparative "Cross-Market" entities.

The data reveals that NYC is not a single market, but rather a collection of disparate micro-economies.

1. **Manhattan** operates as an elite, inelastic premium market. Occupancy remains high regardless of exorbitant price points, driven by corporate travel and international tourism.
2. **Brooklyn** acts as the high-growth transitional market. It offers the highest volume of high-quality, highly-rated residential properties, appealing to domestic tourists seeking authentic neighborhood experiences.
3. **The Bronx and Staten Island** operate functionally as distinct, economy-tier cities. Inventory is highly constrained, properties are more likely to be "Private Rooms" rather than entire homes, and pricing power is severely limited by the distance-to-center geographic penalty.

13. Limitations & Caveats

Analytical transparency requires a candid acknowledgment of data limitations that may impact the margins of our findings:

1. **The Occupancy Proxy Flaw:** Airbnb explicitly obscures true booking data to protect corporate intelligence. Assuming that every blocked calendar day (`available = f`) is a paid booking is the industry standard, but it is inherently flawed. Hosts may block their calendars for personal use, renovations, or regulatory compliance, artificially inflating our `occupancy_rate_365` and `annual_revenue_estimate` calculations.
2. **Survivor Bias:** The scraped dataset only represents properties that were active on the platform on the specific day the scraper ran in 2026. Properties that failed, went bankrupt, or were removed by regulators prior to this date are absent, potentially painting an overly optimistic picture of market success.
3. **Regulatory Blindspots:** New York City has recently implemented draconian short-term rental laws (Local Law 18). Many listings lack verified license numbers. The dataset does not cleanly

identify which listings are operating legally vs. illegally, adding extreme regulatory risk to the financial estimates.

14. Future Improvements

Given additional time, computational resources, and cloud infrastructure, this project would be elevated through several key improvements:

1. **Cloud-Native Pipeline Migration:** The current SQLite architecture is excellent for localized analytics but insufficient for enterprise scale. The pipeline should be refactored to utilize a cloud data warehouse (e.g., Google BigQuery or Snowflake) utilizing Apache Airflow for scheduled, incremental ingestion of the monthly Inside Airbnb updates.
2. **Type 2 Slowly Changing Dimensions (SCD):** The current Dim Host table overwrites data. In a future iteration, a Type 2 SCD model should be implemented (utilizing valid from and valid to timestamps) to track how a host's "Superhost" status changes over historical timeframes, enabling exact point-in-time historical reporting.
3. **Advanced Predictive Modeling:** With the feature engineering complete, a Gradient Boosting model (XGBoost) should be deployed to predict optimal nightly prices based on property attributes, real-time seasonality, and real-time geographic distance.

15. Reflection

Executing this end-to-end data pipeline required balancing deep technical engineering with high-level business storytelling.

Prioritization and Trade-offs: The most significant trade-off made during this project was prioritizing structural data cleanliness over advanced machine learning. A deliberate choice was made to spend hours handling text artifacts, null imputation, and relational key mapping. While this delayed the gratification of building predictive models, it ensured that the final SQLite database was completely bulletproof.

Key Lessons Learned: The primary lesson learned is that real-world data is infinitely messier than academic datasets. Discovering that calendar prices were stored as text (\$1,200.00) and that unrated properties presented as NAN required defensive programming mentalities. Building the automated `AirbnbAutomationPipeline` class underscored the necessity of robust logging; without an audit trail, pipeline failures in a production environment are nearly impossible to trace. Ultimately, this project reinforced that brilliant visualizations are only as valuable as the invisible data engineering that powers them.

Appendix A: AI Usage Disclosure

In adherence to transparency guidelines, the following discloses the use of AI assistance (ChatGPT / Gemini) during the completion of this project:

- **Assisted Sections:** AI was utilized heavily during the **Data Engineering (Section 3)** and **EDA (Section 4)** phases of the Jupyter Notebook.
- **Key Utilization:** * *Syntax Generation:* Assisting with the exact Pandas syntax for complex `groupby` aggregations and datetime parsing.
 1. *Data Modeling:* Generating the SQL logic to construct the Star Schema and assisting in the programmatic generation of the `Dim_Date` table array.

2. *Visualization Styling*: Recommending Matplotlib and Seaborn parameters (`palette`, `figsize`, `alpha`) to ensure visually appealing and professional chart outputs.
 3. *Debugging*: Identifying and resolving `KeyError` exceptions when columns were dropped during DataFrame merges, specifically during the integration of the calendar pricing loops.
- **Validation Notes**: All AI-generated code was manually executed, validated against the raw CSV data to ensure mathematical accuracy, and refined to meet the specific business rules of the New York City dataset. AI was not used to fabricate findings; it acted purely as a co-pilot for code syntax and structural formatting.